

React Components Hands-On Lab — Solution

Building the Student Management Portal ("StudentApp") with Home, About, and Contact Components

Objectives

- Explain React components
- Identify the differences between components and JavaScript functions
- Identify the types of components
- Explain class components
- Explain function components
- Define the component constructor
- Define the render() function

In this hands-on lab, you will learn how to:

- Create a class component
- Create multiple components
- Render a component

Prerequisites

- Node.js
- NPM
- Visual Studio Code

Task

Create a React app for a Student Management Portal named "StudentApp" and create a component named Home that displays the message "Welcome to the Home page of Student Management Portal". Create another component named About that displays the message "Welcome to the About page of the Student Management Portal". Create a third component named Contact that displays the message "Welcome to the Contact page of the Student Management Portal". Call all three components from App.js.

Solution — Step by Step

Step 1: Create the React project "StudentApp"

Open a terminal in Visual Studio Code and run the following command to scaffold the project:

```
npx create-react-app StudentApp
```

Once the setup finishes, navigate into the project folder:

```
cd StudentApp
```

Step 2: Create a Components folder and the Home component

Under the src folder, create a new folder named "components". Inside it, add a new file named Home.js.

Step 3: Write the Home component

Add the following code in Home.js:

```
import React, { Component } from 'react';

class Home extends Component {
  render() {
    return (
      <div>
        <h1>Welcome to the Home page of Student Management Portal</h1>
      </div>
    );
  }
}

export default Home;
```

Step 4: Create the About component

Under the same components folder, add another file named About.js.

Step 5: Create the Contact component

Repeat the same steps to create the About and Contact components. Add About.js and Contact.js inside the components folder, following the same pattern used for Home, but with their own messages, as shown below.

About.js:

```
import React, { Component } from 'react';

class About extends Component {
  render() {
    return (
      <div>
        <h1>Welcome to the About page of the Student Management Portal</h1>
      </div>
    );
  }
}

export default About;
```

Contact.js:

```
import React, { Component } from 'react';

class Contact extends Component {
  render() {
    return (
      <div>
        <h1>Welcome to the Contact page of the Student Management Portal</h1>
      </div>
    );
  }
}
```

```
    }  
  }  
  
  export default Contact;
```

Step 6: Edit App.js to invoke Home, About, and Contact

Update App.js to import and render all three components:

```
import React from 'react';  
import './App.css';  
import Home from './components/Home';  
import About from './components/About';  
import Contact from './components/Contact';  
  
function App() {  
  return (  
    <div className="App">  
      <Home />  
      <About />  
      <Contact />  
    </div>  
  );  
}  
  
export default App;
```

Step 7: Run the application

In the command prompt, navigate into the StudentApp folder (if not already there) and run:

```
cd StudentApp  
npm start
```

Step 8: View the application in the browser

Open a browser window and type the following in the address bar:

```
http://localhost:3000
```

The page will display all three messages one below the other: the Home message, the About message, and the Contact message, confirming that all three components are rendering correctly.

Result

A React application named "StudentApp" was successfully created with three separate class components — Home, About, and Contact — each displaying its own welcome message. All three components were imported and rendered together inside App.js, and the application was verified by running it locally and viewing it in the browser at <http://localhost:3000>.